

FILTER DESIGN TEST CASE

Band-Pass Filter - Band 38 (2600 MHz TDD) for GPS/GNSS Anti-Jamming Filter
Design Validation and Performance Characterization

DOCUMENT INFORMATION

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1. OBJECTIVE

This document describes the design, simulation, and characterization of a Band-Pass Filter filter targeting Band 38 (2600 MHz TDD) for GPS/GNSS Anti-Jamming Filter. The filter is designed on AIN on Si substrate with a target center frequency of 5625.6 MHz and bandwidth of 156.4 MHz. The objective is to validate that the filter meets the required insertion loss, rejection, and linearity specifications for integration into the GPS/GNSS Anti-Jamming Filter module. This design iteration addresses feedback from the previous revision regarding out-of-band rejection improvements and temperature stability.

2. SCOPE

- Filter Type: Band-Pass Filter
- Frequency Band: Band 38 (2600 MHz TDD)
- Application: GPS/GNSS Anti-Jamming Filter
- Substrate: AIN on Si
- Package: LGA (Land Grid Array)
- Design Tool: Sonnet EM Suite v16
- Design Center: Andover HQ Lab

This specification applies to all variants of the Band-Pass Filter design targeting Band 38 (2600 MHz TDD). Results are used to qualify the design for mass production release.

3. DESIGN PARAMETERS

Center Frequency:	5625.6 MHz
Bandwidth (3 dB):	156.4 MHz
Insertion Loss Target:	≤ 1.5 dB
Return Loss Target:	≥ 12.0 dB
Out-of-Band Rejection:	≥ 27.0 dB
Isolation (Duplexer):	≥ 28.0 dB
Q Factor:	6671
Temperature Coefficient:	-13.0 ppm/°C
Die Size:	1.0 x 2.0 mm
Wafer Size:	6-inch
Number of Resonators:	10
Electrode Material:	Al

Passivation: SiO2/Si3N4 stack

4. TEST PROCEDURE

1. Open Sonnet EM Suite v16 and load the Band-Pass Filter template project.
2. Define the substrate stack: AlN on Si with specified layer thicknesses.
3. Set design targets: center freq 5625.6 MHz, BW 156.4 MHz.
4. Run initial EM simulation with 2D mesh density of 20 cells/wavelength.
5. Optimize resonator geometry using gradient descent (target IL < 1.5 dB).
6. Verify coupling coefficients between resonator stages.
7. Run 3D FEM simulation to account for package parasitics (LGA (Land Grid Array)).
8. Export GDS-II layout for mask fabrication.
9. Fabricate prototype wafers (6-inch, AlN on Si).
10. Perform on-wafer probing using Keysight N5227B PNA.
11. Measure S-parameters (S11, S21, S12, S22) from 100 MHz to 16877 MHz.
12. Compare measured results against simulation and specification limits.
13. Perform temperature cycling (-40°C to +85°C) and re-measure.
14. Document results and generate summary report.

5. ACCEPTANCE CRITERIA

- Insertion loss at center frequency shall be <= 1.5 dB
- Return loss in passband shall be >= 12.0 dB
- Out-of-band rejection at +/- 235 MHz offset >= 27.0 dB
- Group delay variation across passband shall be <= 10 ns
- Temperature drift over -40°C to +85°C shall be <= 9.1 MHz
- Power handling shall be >= +28 dBm at 1 dB compression
- ESD tolerance >= 1000 V (HBM)
- Die yield on qualification lot >= 92%

6. TEST RESULTS

Measurement Date: 2024-04-08
Devices Tested: 38
Insertion Loss (meas): 1.18 dB
Return Loss (meas): 14.4 dB
Rejection (meas): 30.3 dB
Isolation (meas): 33.0 dB
Q Factor (meas): 6671
Group Delay Variation: 1.1 ns
Power Handling: +33 dBm
Die Yield: 94.1%

Result Classification: FAIL

7. OBSERVATIONS AND FINDINGS

- a) The Band-Pass Filter design demonstrated below-target performance for Band 38 (2600 MHz TDD).
- b) Insertion loss met target specification.
- c) Out-of-band rejection exceeded the minimum requirement.
- d) Slight frequency drift observed at +85°C; TC layer thickness may need adjustment.
- e) Batch-to-batch reproducibility confirmed over 3 wafer lots.

8. CORRECTIVE ACTIONS

- 1. Optimize resonator geometry to reduce insertion loss by ~0.3 dB.
- 2. Re-run EM simulation with refined mesh for better accuracy.
- 3. Escalate to advanced materials team for alternative piezoelectric stack.

9. SIGN-OFF

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